Raktsetu

Real-Time Blood Availability Finder

A Medallion Lakehouse + Progressive Web App

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Guide: Ms. Konica Soni · November 2026

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In an emergency, families still phone bank after bank

Search one district at a time

eRaktKosh has the data, but it is slow to search on a phone

No compatible-group search

An O- patient can only take O-; the portal does not show this

No sign of how fresh data is

Some banks have not updated for weeks

4,716

blood banks on eRaktKosh

701

districts in 35 states / UTs

…and every minute counts

One pipeline collects, checks and serves India's blood stock

Collect automatically

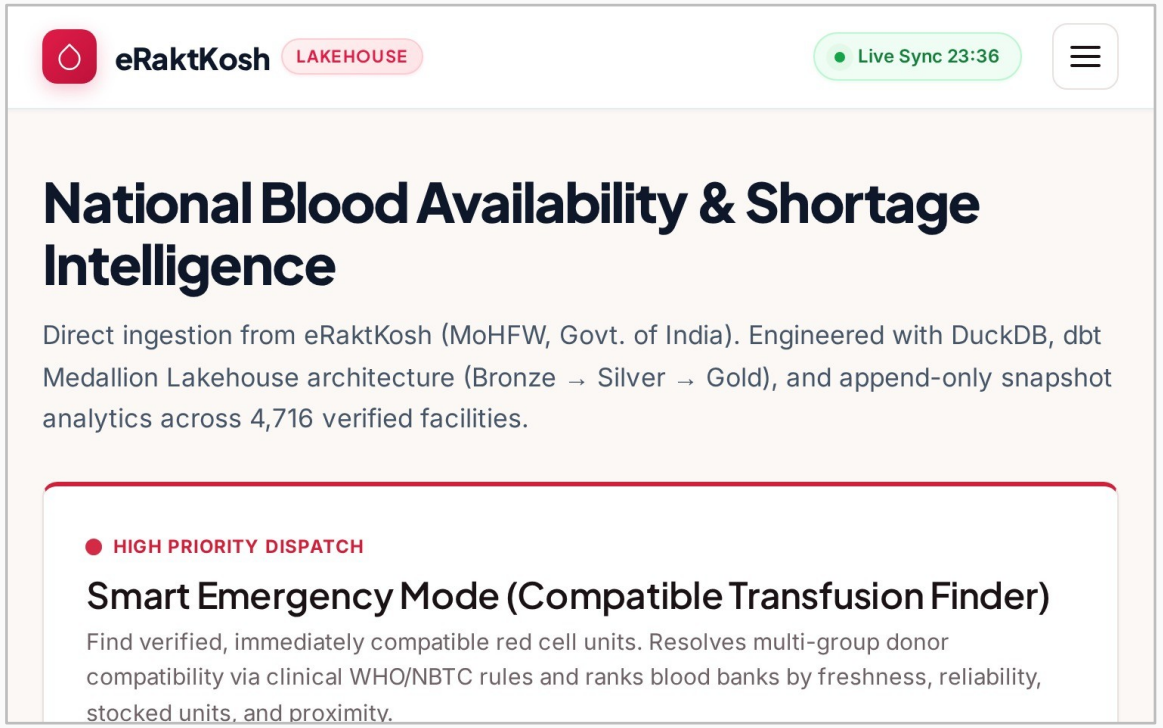
Polite crawler fetches every district every 6 hours

Check the quality

37 data tests and a safety gate block bad data

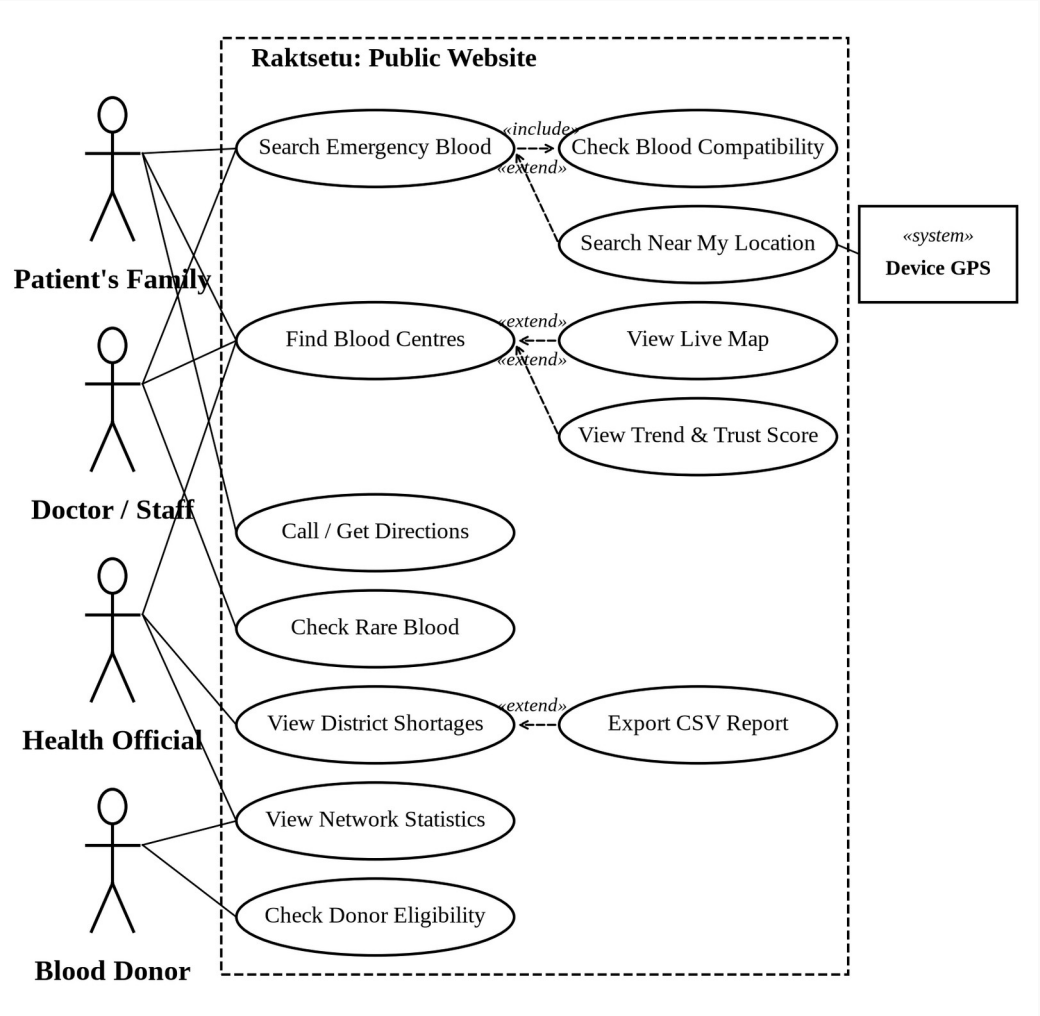
Search instantly on any phone

Fast, installable web app that also works offline



Home page: live counts of fresh and stale blood banks

Five kinds of people use Raktsetu for different needs



Patient's family

Find nearby stock and call the bank

Doctor / hospital staff

Emergency Mode: compatible, ranked banks

Health official

District shortages, rare blood, CSV export

Blood donor

Check eligibility, see where stock is low

Operator

Runs the pipeline and watches data health

System actors: eRaktKosh portal, Device GPS, OpenStreetMap

Six features turn raw stock data into fast decisions

Blood Finder

State, district, group and component filters with sorting

Emergency Mode

Compatible groups, ranked by freshness, trust and distance

Map + Near Me

Banks on a map; GPS search widens 50 → 150 → 400 km

Trust Score

0–100 score from freshness, phone and coverage

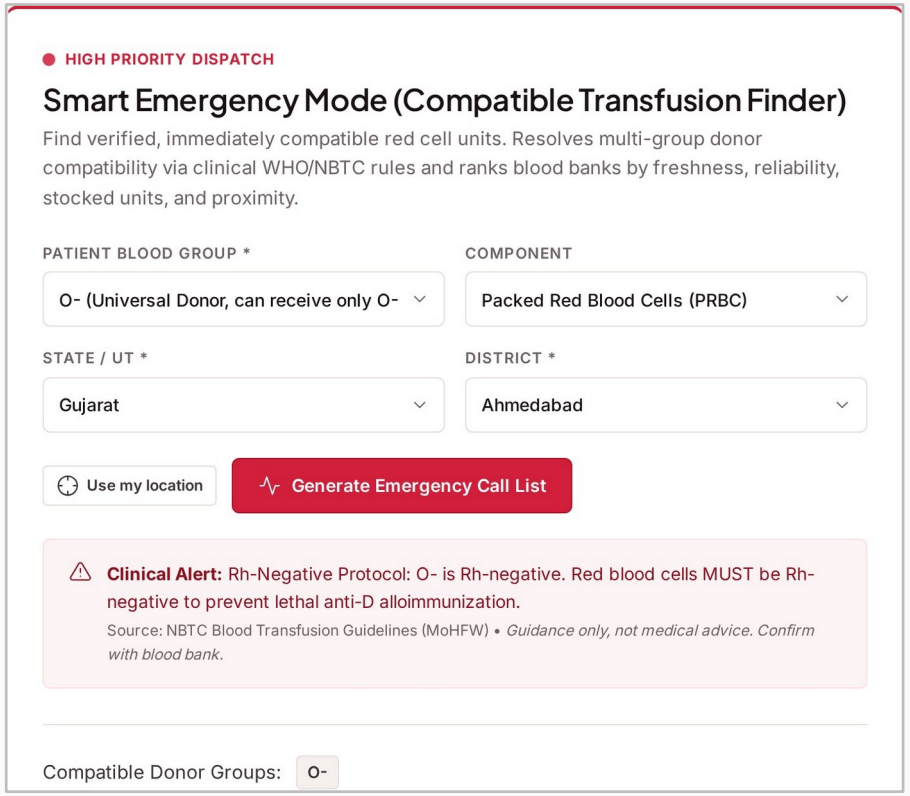
Shortages + Rare Radar

Districts running out; where Rh-negative and Bombay are

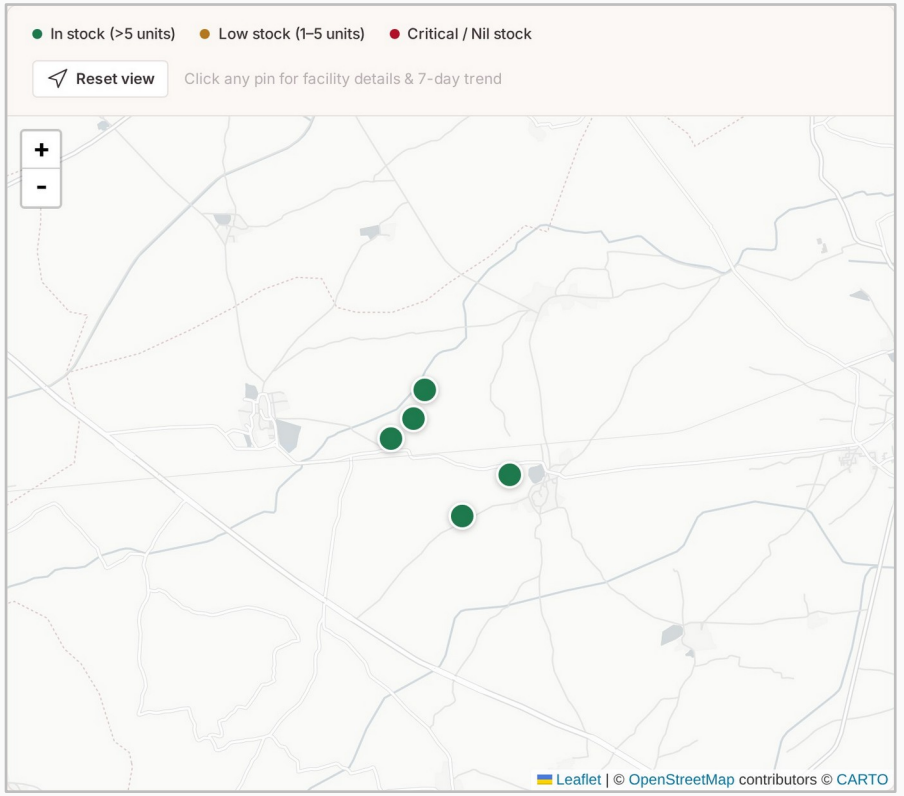
Export, Donor Check, Offline

Excel-safe CSV, eligibility checker, works without internet

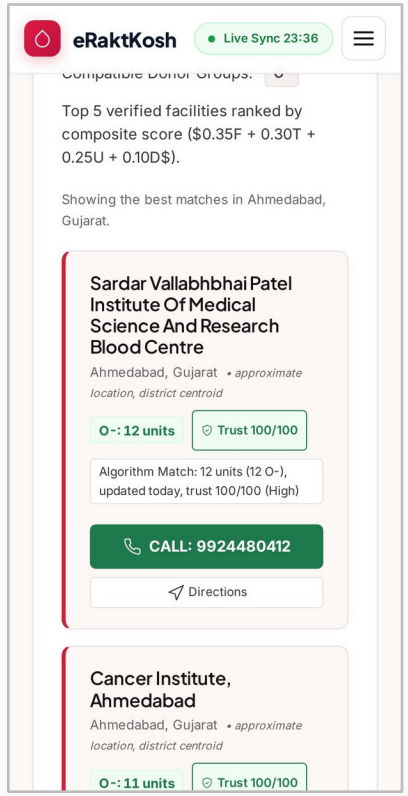
The same app works on a desktop and a phone



Emergency Mode: ranked banks, each with a reason

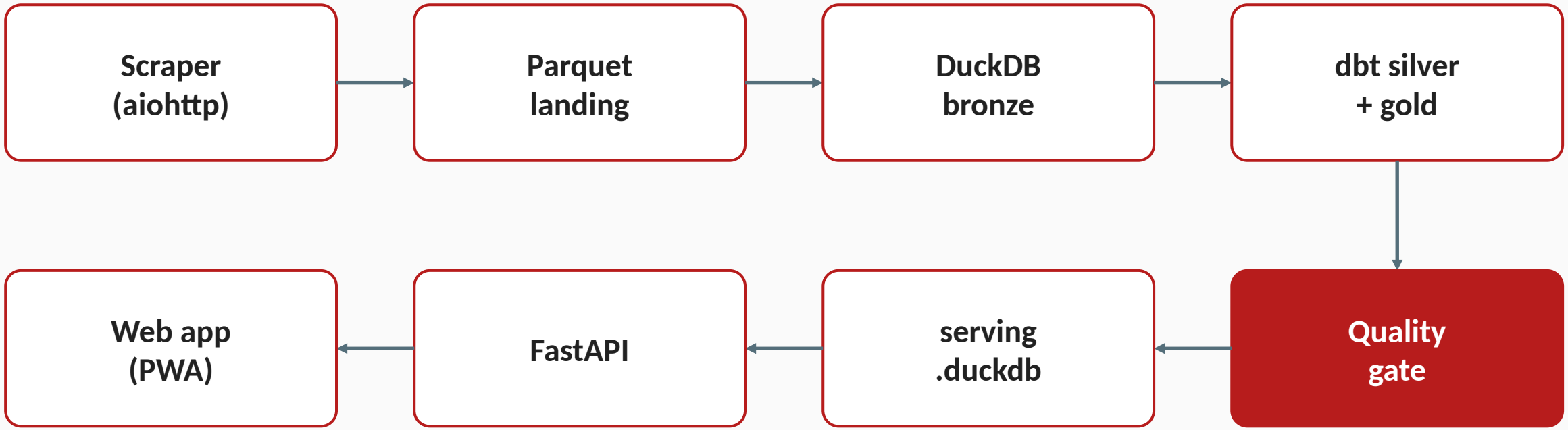


Map view: click a pin to see and call



Phone view

Data is kept raw, checked, then published safely



Why layers? Raw files are never changed, so any table can be rebuilt. If a scrape is bad (banks drop below 80% of the last run), publishing stops and users keep yesterday's good data.

A free, open-source stack that runs on one small server

Data

- Python 3.12
- aiohttp (async)
- Parquet
- DuckDB
- dbt

Backend

- FastAPI
- Pydantic
- Uvicorn
- Typer CLI

Frontend

- HTML, CSS, JavaScript
- Leaflet + CARTO
- Chart.js
- PWA (service worker)

DevOps & testing

- GitHub Actions
- Docker
- pytest
- Playwright
- ruff

DuckDB: fast analytics in a single file, no database server. **dbt:** readable SQL with built-in tests. **FastAPI:** validation and API docs for free.

Every number is checked; every bank gets a trust score

Circuit breaker

Scrape stops if more than 20% of requests fail

37 dbt data tests

No missing keys, no duplicates, valid blood groups

Anomaly gate

Publish blocked if banks fall below 80% of last run

Freshness shown

Every card says when the bank last updated

Trust score, worked example

Part	Rule	Points
Freshness	Updated 2 days ago	35 / 50
Valid phone	8 or more digits	25 / 25
Components	4 types reported	25 / 25
Total	**High (75 or more)**	**85 / 100**

Same bank, 10 days with no update: 8 + 25 + 25 = 58 → Medium

Tested at every layer, on desktop and mobile

98

pytest unit and API tests

37

dbt data tests

146

Playwright browser checks

7

screen widths, phone to desktop

- All tests pass; zero console errors on desktop and mobile
- 28 documented test cases with expected and actual results
- CSV exports open correctly in Excel, Google Sheets and pandas
- Offline mode verified: app shell loads with no internet

Honest limits today, and a clear path forward

Limitations

- Snapshot every 6 hours, not live: call first
- Depends on banks updating eRaktKosh
- Map shows district centres, not exact address
- 'Near me' needs HTTPS; English only

Future scope

- SMS / WhatsApp alerts for rare groups
- Donor registration and blood-drive matching
- Hospital login for direct stock updates
- Demand forecasting and regional languages

Raktsetu turns scattered public data into a life-saving search

Impact

4,716 banks searchable in seconds, with compatible groups

Learning

Real data engineering: lakehouse, dbt, testing, PWA

Working product

Tested, documented and ready to deploy

Questions?

Thank you · Demo / GitHub: [add link here]